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Recommended Design Rules and Strategies for BGA Devices User Guide (UG1099)

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Introduction

AMD Versal™ architecture, AMD UltraScale™ architecture, 7 series, AMD Virtex™ 6, and AMD Spartan™ 6 devices come in a variety of packages that are designed for maximum performance and maximum flexibility. Four pitch sizes are available for these packages: 1.0 mm, 0.92 mm, 0.8 mm, and 0.5 mm. In general, as the pitch size decreases, the challenges for PCB routing increase as there is less room to route traces and vias between package balls. This guide illustrates various methods for successful design regardless of pitch size.

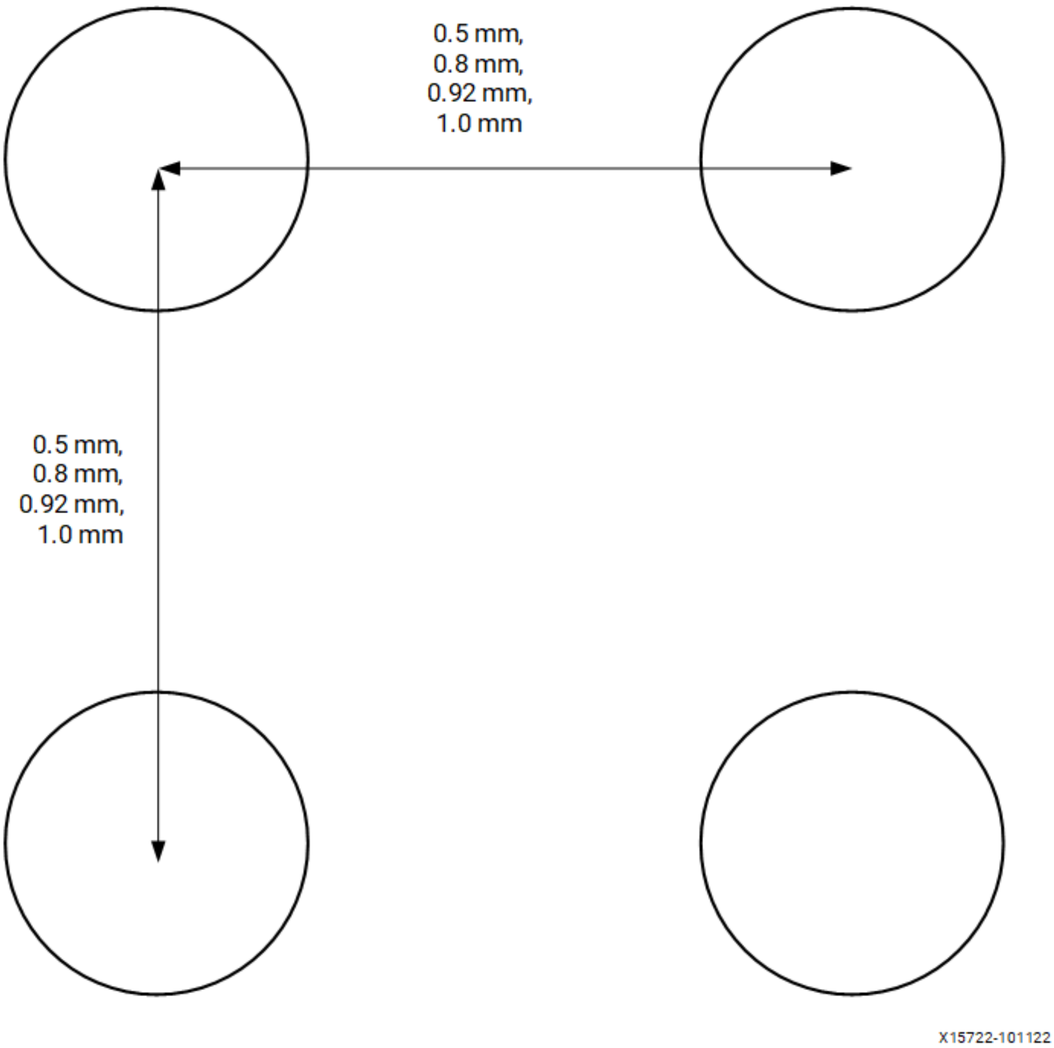
✎ Note: Throughout this guide, various specifications and estimates are given regarding PCB pricing, costs, and technology. As PCB manufacturing technology is constantly advancing, it is highly advised to consult with your PCB manufacturer to fully understand their capabilities regarding the information presented here.

General BGA and PCB Layout Overview

The primary factor that determines the intricacy of BGA routing is the pitch size. In addition, factors such as the size of the BGA array, type of solder masks used, and layer count requirements also play crucial factors.

Pitch size is defined as the distance between consecutive balls on a BGA package, measured from center-to-center, as shown in the following diagram.

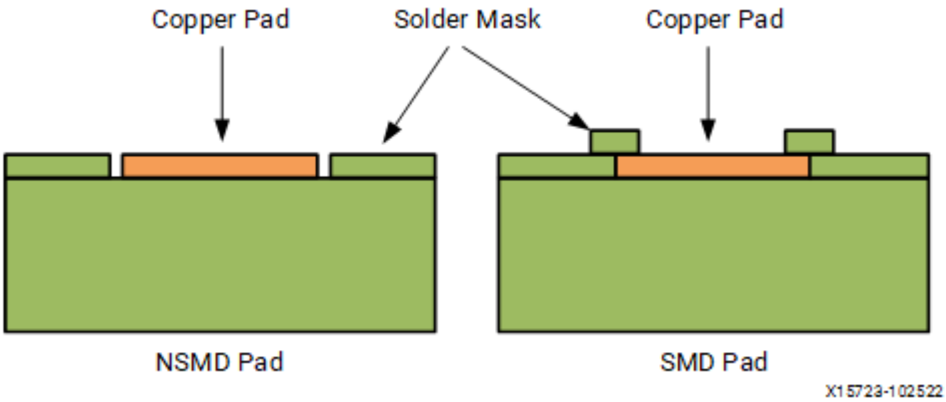
Figure: Definition of Pitch Size



BGA Landing Pads

AMD recommends using non-solder mask-defined (NSMD) copper BGA landing pads for optimum board design. NSMD pads are pads that are not covered by any solder mask, as opposed to Solder Mask Defined (SMD) pads in which a small amount of solder mask covers the pad landing. The following figure illustrates the difference between NSMD and SMD pads.

Figure: NSMD and SMD Pads



Layer Count Estimation and Optimization

Layer Count Estimation

A quick way to estimate the number of signal routing layers required to fully break out signal pins from the FPGA would be to use the following equation:

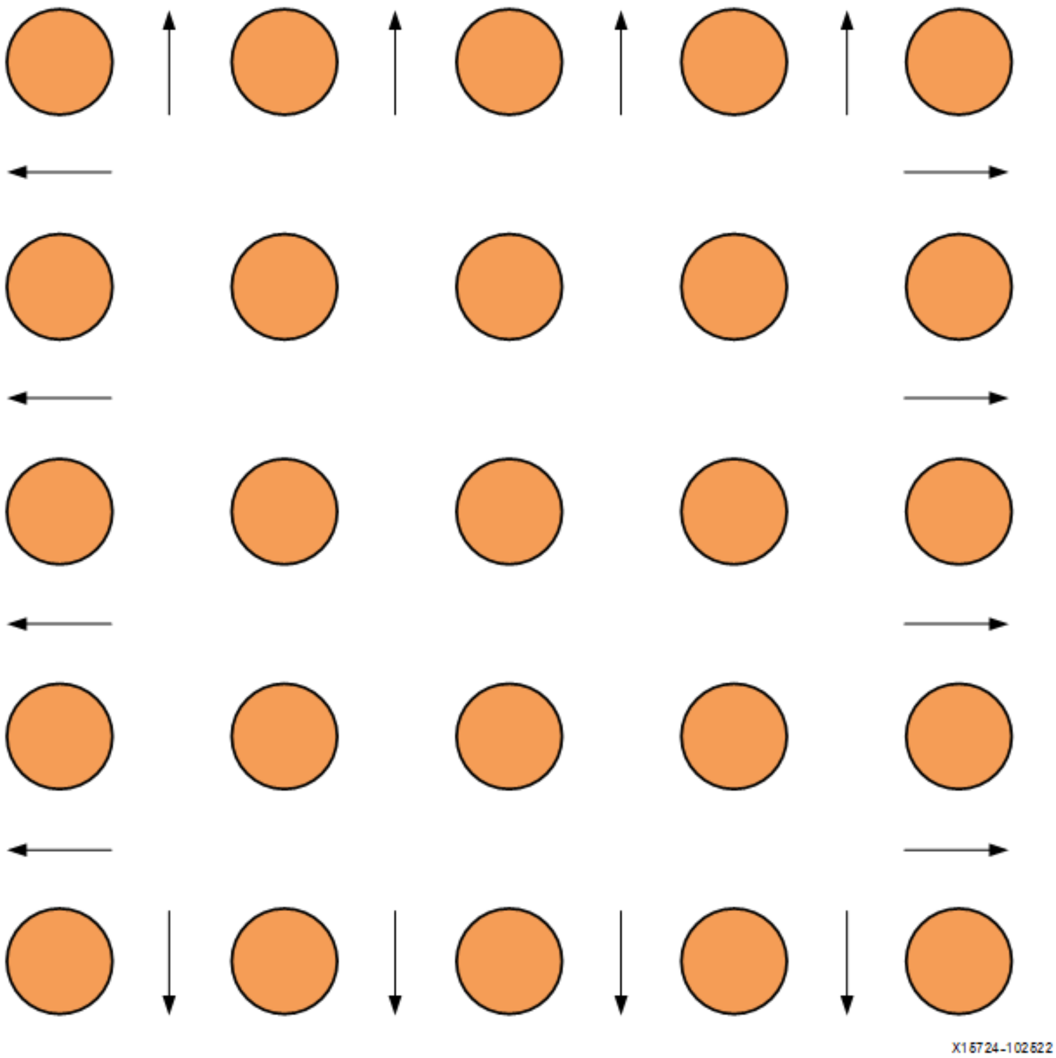
Figure: Layers

$$Layers = \frac{Signals}{Routing\ Channels\ x\ Routes\ Per\ Channel}$$

For AMD FPGAs, MPSoCs/RFSoCs, and adaptive SoCs, the quantity of signals is approximately 60% of the number of BGA balls. The other 40% are power and ground signals that are most often routed directly down to their own dedicated planes by vias. The equation assumes full I/O utilization. If fewer I/Os are used, the number of signals to route goes down accordingly.

Routing channels are the number of available routing paths out of the BGA area (the number of BGA pins on one side minus one, times four sides). The following figure shows a 5x5 grid with sixteen total routing channels (four routing channels per side times four sides).

Figure: Definition of Routing Channel (16 Total Routing Channels Shown)



Routes per channel is either one or two, depending on whether one or two signals are routed between BGA pads. The approximate number of signal layers required to fully route out an AMD FPGA or adaptive SoC are shown in the following table.

Table: Approximate Signal Layers per no. of Package Pins

BGA Pins	Ball Pitch (mm)	Routing Channels	Estimated Signal Layers Required with All Available I/Os Routed	
			One Trace Per Routing Channel	Two Traces Per Routing Channel

BGA Pins	Ball Pitch (mm)	Routing Channels	Estimated Signal Layers Required with All Available I/Os Routed	
			One Trace Per Routing Channel	Two Traces Per Routing Channel
196	0.5	52	2	1
196	1.0	52	2	1
225	0.8	56	2	1
236	0.5	56	3	1
238	0.5	56	3	1
256	1.0	60	3	1
324	0.8	68	3	1
400	0.8	76	3	2
484	0.8/1.0	84	3	2
485	0.8	84	3	2
494	0.5	84	4	2
530	0.5	88	4	2
625	0.8	96	4	2
676	1.0	100	4	2
784	0.8/1.0	108	4	2
900	1.0	116	5	2
1024	0.92	124	5	2
1154	1.0	132	5	3
1155	1.0	132	5	3

BGA Pins	Ball Pitch (mm)	Routing Channels	Estimated Signal Layers Required with All Available I/Os Routed	
			One Trace Per Routing Channel	Two Traces Per Routing Channel
1156	1.0	132	5	3
1157	1.0	132	5	3
1365	0.92	144	6	3
1369	0.92	144	6	3
1517	1.0	152	6	3
1596	0.92	156	6	3
1759	1.0	164	6	3
1760	0.92/1.0	164	6	3
1761	1.0	164	6	3
1923	1.0	172	7	3
1924	1.0	172	7	3
1925	1.0	172	7	3
1926	1.0	172	7	3
1927	1.0	172	7	3
1928	1.0	172	7	3
1930	1.0	172	7	3
2104	1.0	180	7	4
2197	0.92	184	7	4
2377	1.0	188	8	4

BGA Pins	Ball Pitch (mm)	Routing Channels	Estimated Signal Layers Required with All Available I/Os Routed	
			One Trace Per Routing Channel	Two Traces Per Routing Channel
2577	1.0	200	8	4
2785	0.92	208	8	4
2892	1.0	212	8	4
3340	0.92	228	9	4
3824	1.0	244	9	5
4072	1.0	252	10	5
5601	0.92	296	11	6

Layer Count Optimization

Versal architecture, UltraScale architecture, 7 series, Virtex 6, and Spartan 6 device packages have full matrices of solder balls. The true number of layers required for effective routing of these packages is dictated by a variety of factors, including:

- BGA size (quantity of pins)
- Pad size, pad pitch, and trace width
- Fixed pinouts
- Back Drilling
- Fabrication Technologies

BGA Size

The quantity of pins in a BGA indicates the number of signals to route. Because of physical space constraints, the quantity of signals required to route is proportional to the amount of signal layers required.

Pad Size, Pad Pitch, and Trace Width

The pad size and pitch determine the available space between adjacent balls for signal escape. Based on the chosen trace width, one or two signals can be routed between adjacent pads. If one signal escapes between adjacent pads, then one signal row can be routed on a single metal layer. The exception to this is the outermost row, which allows two routes per layer.

To facilitate routing in the ball grid area, necking down the trace width in the critical space between the BGA pads/vias (the breakout area) is allowable. This then allows for two signal rows to be routed on a single metal layer (or three if routing the outermost row). The traces can then be widened after they escape the breakout area. Changes in width over very short distances can cause small impedance changes. Validate these issues with the board vendor and signal integrity engineers responsible for the design.

Fixed Pinouts

AMD FPGA and adaptive SoC pinouts are designed with maximum flexibility in mind. However, certain FPGA/adaptive SoC signals, such as JTAG, transceiver inputs and outputs, and memory controller signals (among others) have fixed locations, which means routing of these signals is limited compared to other signals that can be swapped as needed. Fixed locations lead to layout trade-offs that can have an impact on the number of required signal layers.

Back Drilling

Back drilling is the technique is which unused via stubs have their metal drilled away to remove the potential for the stubs to cause reflections which can cause signal integrity problems. Typically, back drilling can prevent the ability to route more than one signal in-between pads and vias due to manufacturability concerns. Always consult with the PCB manufacturer regarding the ability to back drill before beginning and layout activity.

Fabrication Technologies

Several advanced fabrication technologies can be used to reduce the number of layers required to route a design.

Blind Vias

Unlike through-hole vias, which extend through the entire thickness of the PCB from the top layer all the way to the bottom layer, blind vias connect only an outer layer (either the top or bottom) to one or more inner signal layers without passing completely through the board. This selective connectivity enables designers to free up valuable routing space on the outer layers by avoiding the need to route signals through the entire board thickness. As a result, blind vias help improve signal density and reduce layer count, which can lead to smaller, more compact PCB designs.

Buried Vias

Buried vias are vias that connect only inner layers within the PCB stack-up and do not extend to the top or bottom surface layers. Because they are completely encapsulated inside the PCB, buried vias are not visible or accessible from the outside. Their primary purpose is to enable high-density interconnections between internal layers without occupying valuable surface real estate or interfering with surface-mounted components. Buried vias are particularly useful in multi-layer boards where signal integrity and routing complexity demand precise control of internal layer connections.

Via-in-Pad

Via-in-pad technology places a via directly within a component pad, such as those found on BGA packages. This method eliminates the need for “dog-bone” routing, where a signal trace extends from the pad to a nearby via on the outer layers. By situating the via right in the pad, the signal can travel directly from the component pin down to an inner layer, significantly simplifying the routing process under dense BGAs and other fine-pitch components.

This approach offers several advantages:

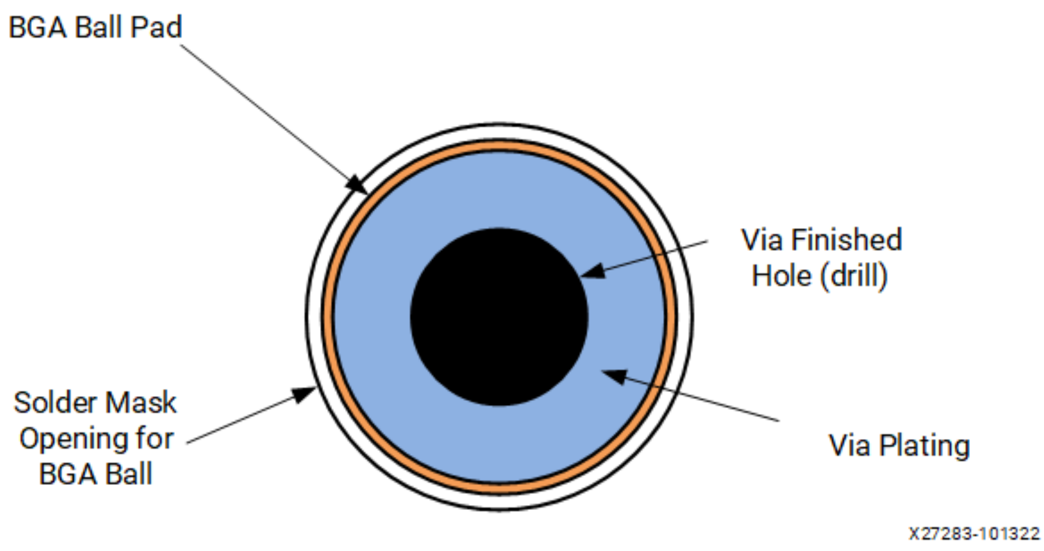
- **Improved Routing Density**

Via-in-pad frees up surface routing space, allowing easier escape routes for signals, especially under BGAs where routing channels are extremely limited.

- Enhanced Signal Integrity
  Because the signal path transitions directly from the pad to an internal layer, the length of the top or bottom layer trace is minimized or eliminated, reducing parasitic inductance and capacitance. This improves impedance control and overall high-frequency performance.
- Better Thermal Performance
  Via-in-pad also helps with heat dissipation from components by providing a direct thermal path to internal or bottom layers.

However, via-in-pad requires specialized manufacturing processes such as via filling and plating to ensure flat, reliable pads suitable for soldering components. The following figure illustrates the mechanical design of a via-in-pad, showing the via embedded within the pad structure.

Figure: Via-In-Pad Structure



Maximum Board Thickness and Aspect Ratio

The maximum board thickness is a function of the minimum drill diameter and aspect ratio, both of which are provided by the PCB manufacturer. A typical aspect ratio of 15:1 indicates that the board can be no thicker than fifteen times the drill diameter. A drill diameter of 10 mils, for example, would lead to a maximum board thickness of 150 mils. Apart from the CP package, AMD recommends finished drill diameters to be 10-15 mils, which translates to an actual drill diameter of about 13-18 mils (plating typically reduces the diameter by about 3 mils). A 10 mil drill would lead to a maximum board thickness of 100 mils for a 10:1 ration, or 150mils for a 15:1 ratio. Advanced manufacturing technologies can support from 17:1 to 22:1 ratio, but at increased costs.

Recommended BGA Ball Pad, Via, and Trace Dimensions for 1.0 mm, 0.92 mm, 0.8 mm, and 0.5 mm Devices

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✎ Note: The figures in this chapter assume that signals are routed from the BGA ball to a via in a dog-bone configuration where the signal from the BGA routes diagonally from the pad to a via on the same routing layer. Via-In-Pad (VIP) technology (see the Via-In-Pad Structure figure in Fabrication Technologies) can be used to place a via directly on top of a BGA pad so it can travel straight down to an inner signal layer.

Recommended BGA Ball Pad and Via Dimensions for 1.0 mm, 0.92 mm, 0.8 mm, and 0.5 mm Devices

The definition of dimensions of FPGA/adaptive SoC ball pads and vias for AMD BGA devices are show in the following figure. The table that follows shows the actual dimensions based on BGA ball pitch.

Figure: BGA Ball and Via Definition of Dimensions

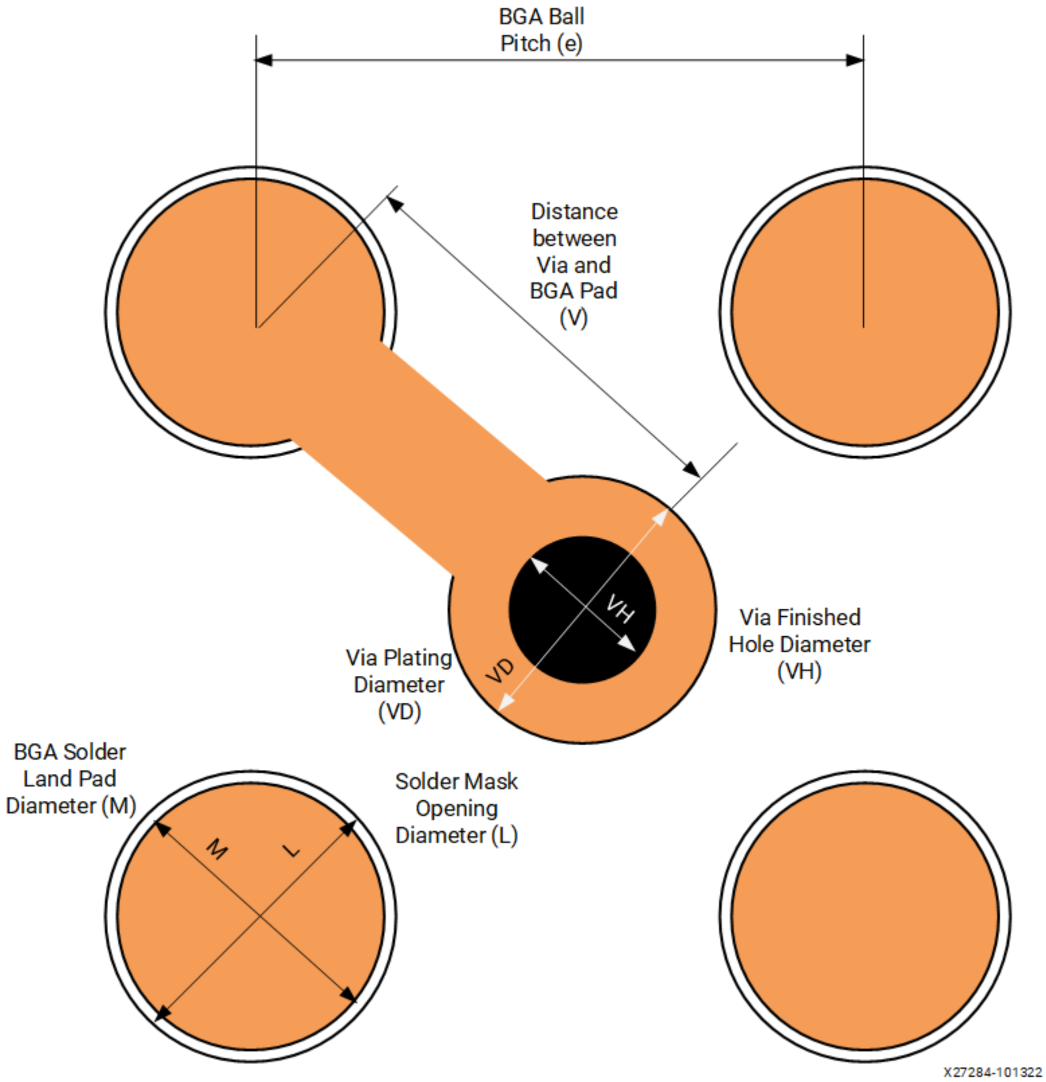


Table: BGA Ball and Via Dimensions

Solder Ball Land Pitch (e)	1.0 mm	0.92 mm	0.8 mm	0.5 mm
Solder Mask Opening Diameter (L)	20.9 mil	20.9 mil	15.7 mil	11.0 mil
	0.53 mm	0.53 mm	0.40 mm	0.28 mm
BGA Solder Land Pad Diameter (M)	19.7 mil	20.0 mil	15.7 mil	10.2 mil
	0.50 mm	0.51 mm	0.40 mm	0.26 mm

Solder Ball Land Pitch (e)	1.0 mm	0.92 mm	0.8 mm	0.5 mm
Via Plating Diameter (VD)	19 mil	19 mil	19 mil	10 mil
	0.48 mm	0.48 mm	0.48 mm	0.254 mm
Via Finished Hole Diameter (VH)	10 mil	10 mil	10 mil	4 mil
	0.25 mm	0.25 mm	0.25 mm	0.10 mm
Distance between BGA Pad and Via)	27.83 mil	25.61 mil	22.27 mil	13.92 mil
	0.70 mm	0.65 mm	0.56 mm	0.35 mm

Recommended Trace Routing between Pads and Vias for 1.0 mm, 0.92 mm, 0.8 mm, and 0.5 mm Devices

The ball pitch and BGA pad/via diameters determine how much space is available to route traces between pads or vias. Standard PCB processes can allow for as low as 3.5 mil trace widths with 3.5 mil spacing. Advanced processes can allow for as low as 2 mil trace widths with 2 mil spacing. Recommended trace routing is shown in the following figure. The table that follows shows the actual BGA/trace routing dimensions.

Figure: BGA/Trace Routing Dimensions

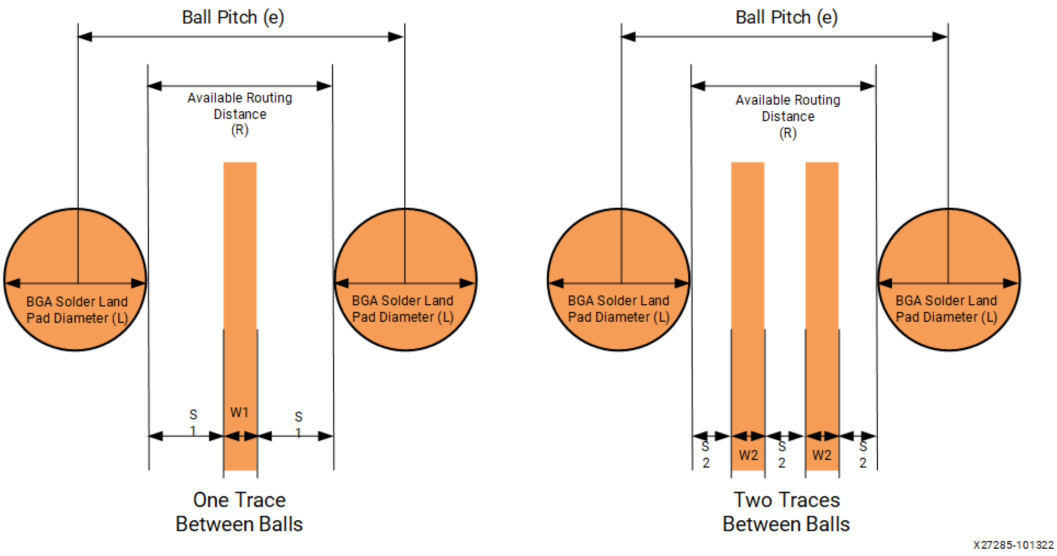


Table: BGA/Trace Routing Dimensions

Solder Ball Land Pitch (e)	39.39 mil 1.0 mm	36.22 mil 0.92 mm	31.40 mil 0.8 mm	19.7 mil 0.5 mm
PCB Solder Land Diameter (L)	19.7 mil	20.0 mil	15.7 mil	10.63 mil
	0.50 mm	0.51 mm	0.40 mm	0.27 mm
Available Routing Distance (R)	19 mil	16 mil	15 mil	9 mil
	0.33 mm	0.40 mm	0.38 mm	0.23 mm
Pad-to-Trace/Trace-to-Trace Spacing for one route between pads (S1)	7.5 mil	6 mil	5 mil	3 [1]
	0.19 mm	0.15 mm	0.13 mm	0.08 mm
Trace thickness for	4	4	4	3

Solder Ball Land Pitch (e)	39.39 mil 1.0 mm	36.22 mil 0.92 mm	31.40 mil 0.8 mm	19.7 mil 0.5 mm
one route between pads (W1)	0.10 mm	0.10 mm	0.10 mm	0.08 mm
Pad-to-Trace/Trace-to-Trace Spacing for two route between pads (S2)	4 mil	3 mil	3 mil	N/A
	0.10 mm	0.08 mm	0.08 mm	
Trace thickness for two route between pads (W2)	3 mil	3 mil	3 mil	N/A
	0.08 mm	0.08 mm	0.08 mm	

1. 3 mil pad-to-trace is not supported by all manufacturers.

Recommended Trace Routing between Vias

One or two traces can be routed in-between vias for 1.0 mm, 0.92 mm, and 0.80 mm pitch devices. It is not practical to route traces in-between vias spaced 0.5 mm apart due to the tight spacing and trace widths required. For those situations, it is recommended to use Via-in-Pad technology for added pitch between vias. Recommended via/trace routing is shown in the following figure. The table that follows shows the actual BGA/trace routing dimensions.

Figure: Via/Trace Routing

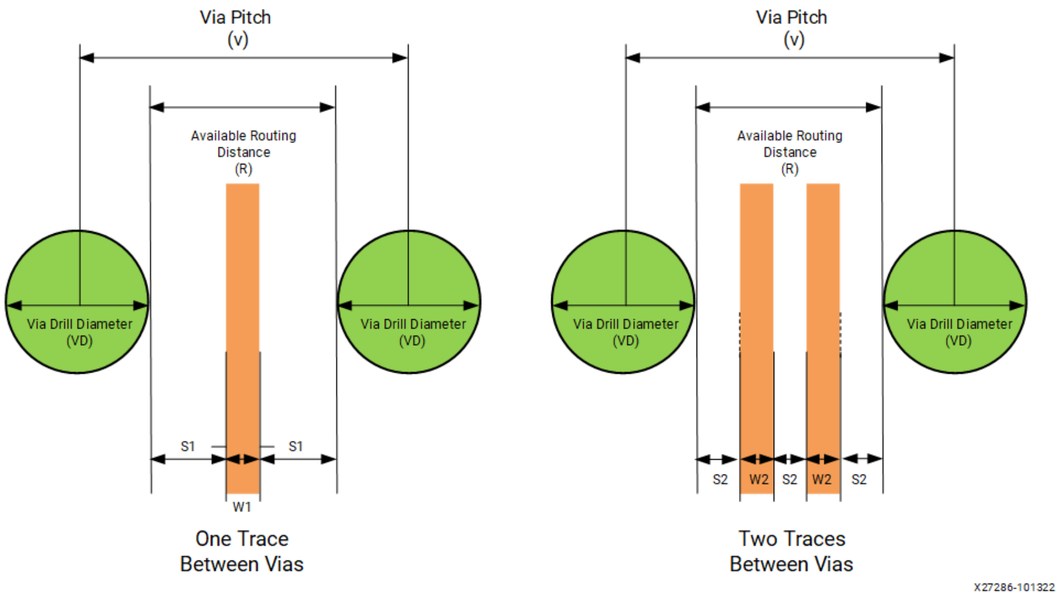


Table: Via/Trace Routing Dimensions

Solder Ball Land Pitch (e)	39.39 mil 1.0 mm	36.22 mil 0.92 mm	31.40 mil 0.8 mm	19.7 mil 0.5 mm
Via Drill Diameter (VD)	10 mil	10 mil	10 mil	4 mil
	0.25 mm	0.25 mm	0.25 mm	0.10 mm
Available Routing Distance (R)	29 mil	26 mil	21 mil	15 mil
	0.74 mm	0.66 mm	0.53 mm	0.38 mm
Via-to-Trace/ Trace-to-Trace Spacing for one route between pads (S1)	12 mil	11 mil	8 mil	5 mil
	0.30 mm	0.28 mm	0.20 mm	0.13 mm

Solder Ball Land Pitch (e)	39.39 mil 1.0 mm	36.22 mil 0.92 mm	31.40 mil 0.8 mm	19.7 mil 0.5 mm
Trace thickness for one route between pads (W1)	4 mil	4 mil	4 mil	4 mil
	0.10 mm	0.10 mm	0.10 mm	0.10 mm
Pad-to-Trace/Trace-to-Trace Spacing for two route between pads (S2)	10 mil	6 mil	5 mil	3 mil [1]
	0.25 mm	0.15 mm	0.13 mm	0.08 mm
Trace thickness for two route between pads (W2)	4 mil	4 mil	3 mil	3 mil
	0.10 mm	0.10 mm	0.08 mm	0.08 mm

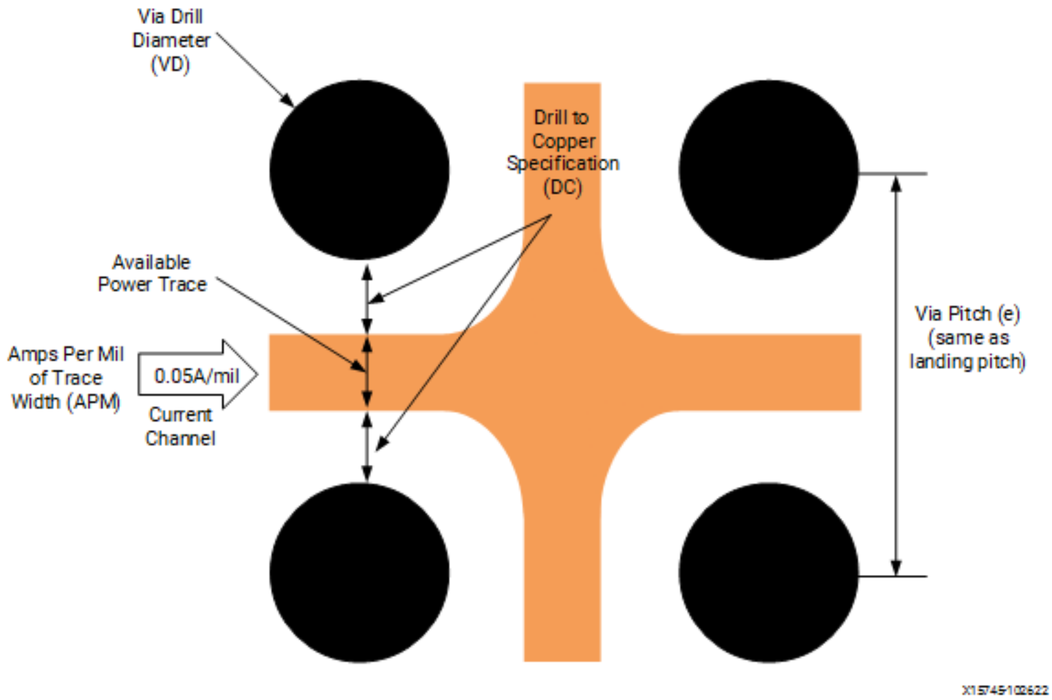
1. 3 mil pad-to-trace is not supported by all manufacturers.

Power Delivery to the FPGA

Power needs must be assessed early in the design phase to assure that there are enough layers and area to provide sufficient power to the BGA balls that require power. Because most of the BGA power pins are located in the center of the BGA area, the path the current travels traverses a myriad of vias in the BGA area. The space between vias can conservatively carry about 0.05A per mil of trace width (for 0.5 oz copper). The trace width between vias is defined by the pitch of the vias (usually the same as the pitch of the BGA), the via drill diameter, and drill-to-copper specification as defined by the fabrication house. The following figure shows how to calculate the amount of current that can pass through each via channel. Ensure that the power planes are wide enough and encompassing enough to supply the needed amperage to the BGA power balls. The following equation can be used to calculate the current per channel:

$$\begin{aligned}&Current\ Per\ Channel\\&\qquad = (ViaPitch - ViaDrillDiameter\\&\qquad - 2xDrillToCopperSpecification)xAmpsPerUnitTraceWidth\end{aligned}$$

Figure: Power Delivery within BGA Area (0.5 oz Copper)



The following table shows current per channel values for 0.8 mm and 1.0 mm pitch devices. Because of the very fine pitch of 0.5 mm devices, it is not possible to route in-between standard vias. Micro-vias under the BGA pads are recommended for 0.5 mm devices in order to reach the power planes

Table: Current Per Channel Calculation for 0.8 mm, 0.92 mm, and 1.0 mm Devices

	0.8 mm Pitch	0.92 mm Pitch	1.0 mm Pitch
Via Pitch	31.40 mil	36.22 mil	39.37 mil
Via Drill Diameter	10 mil	10 mil	10 mil

	0.8 mm Pitch	0.92 mm Pitch	1.0 mm Pitch
Drill to Copper Specification [1]	8 mil	8 mil	8 mil
Amps per Unit Trace Width (ATW), 0.5 oz Cu	0.05 A	0.05 A	0.05 A
Amps per Unit Trace Width (ATW), 1.0 oz Cu	0.075 A	0.075 A	0.075 A
Current per Channel, 0.5 oz Cu	0.27 A	0.51 A	0.67 A
Current per Channel, 1.0 oz Cu	0.41 A	0.77 A	1.00 A

1. Drill-to-Copper of 8 mils is considered standard. Between 6.5 and 8 mil are considered advanced processes.

Sample Breakouts for 1.0 mm, 0.92, and 0.8mm Pitch Devices

The following figures show two example layouts showing the BGA routing breakout areas of a representative 1.0 mm/0.92 mm/0.80 mm pitch AMD device.

Sample Breakout with One Route between BGA Balls and Vias

Figure: Top Layer Breakout (1.0 mm/0.92 mm/0.8 mm), One Route between BGA Balls

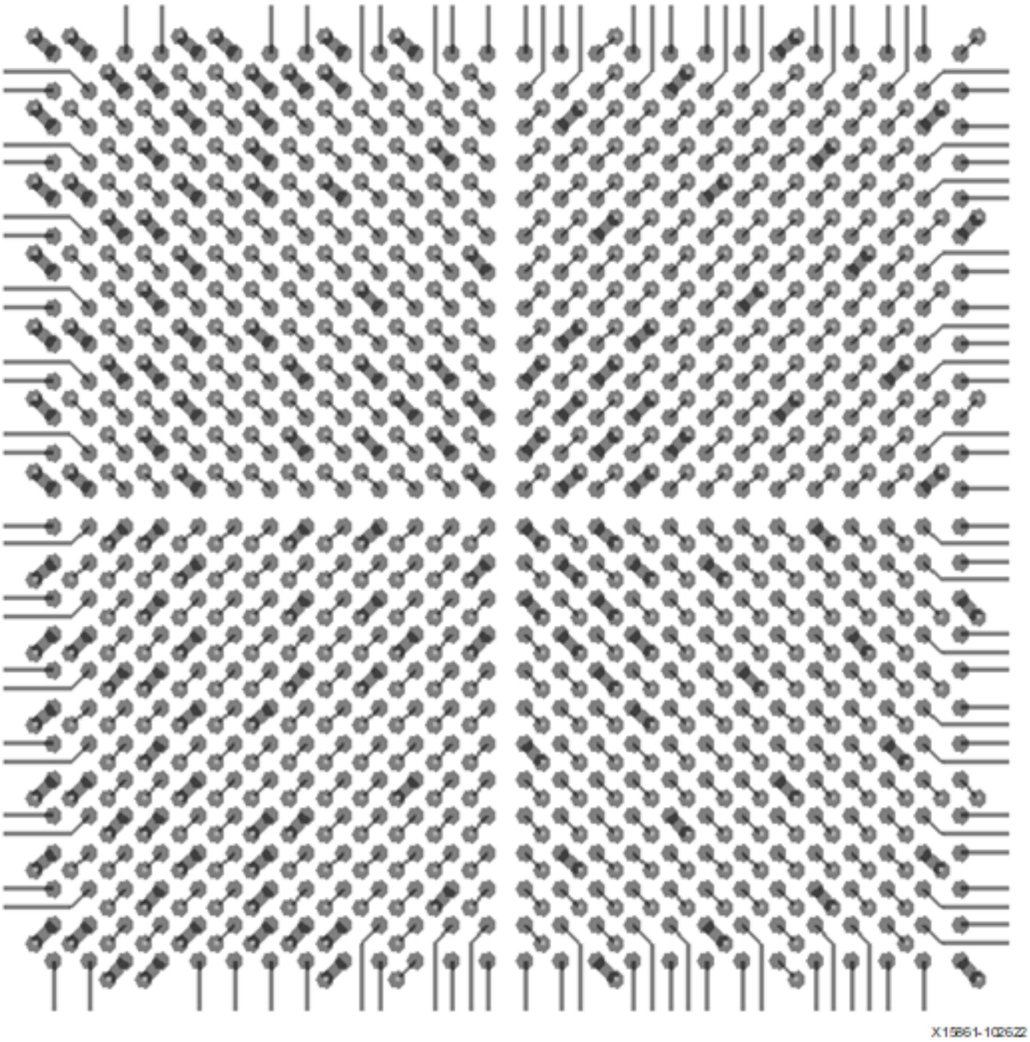


Figure: Inner Signal Layer One Breakout (1.0 mm/0.92 mm/0.8 mm), One Route between Vias

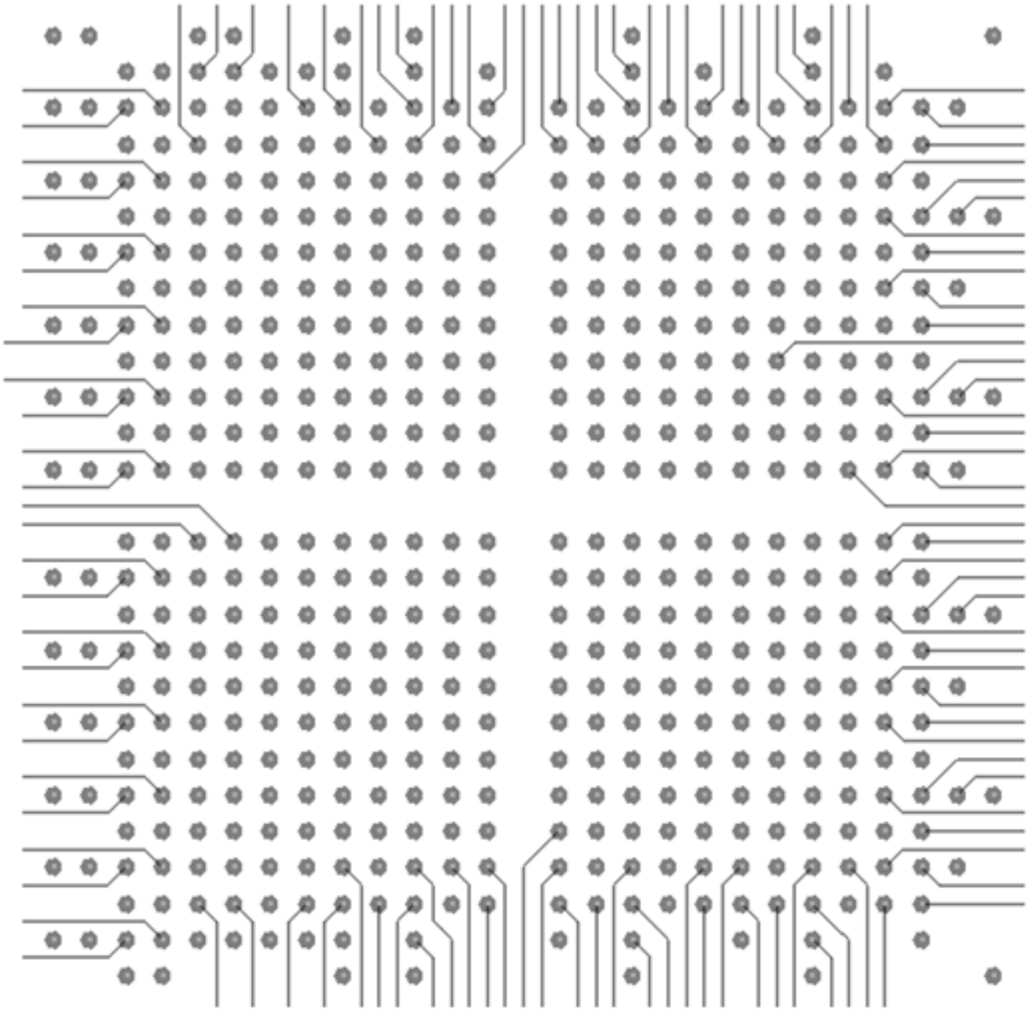


Figure: Inner Signal Layer Two Breakout (1.0 mm/0.92 mm/0.8 mm), One Route between Vias

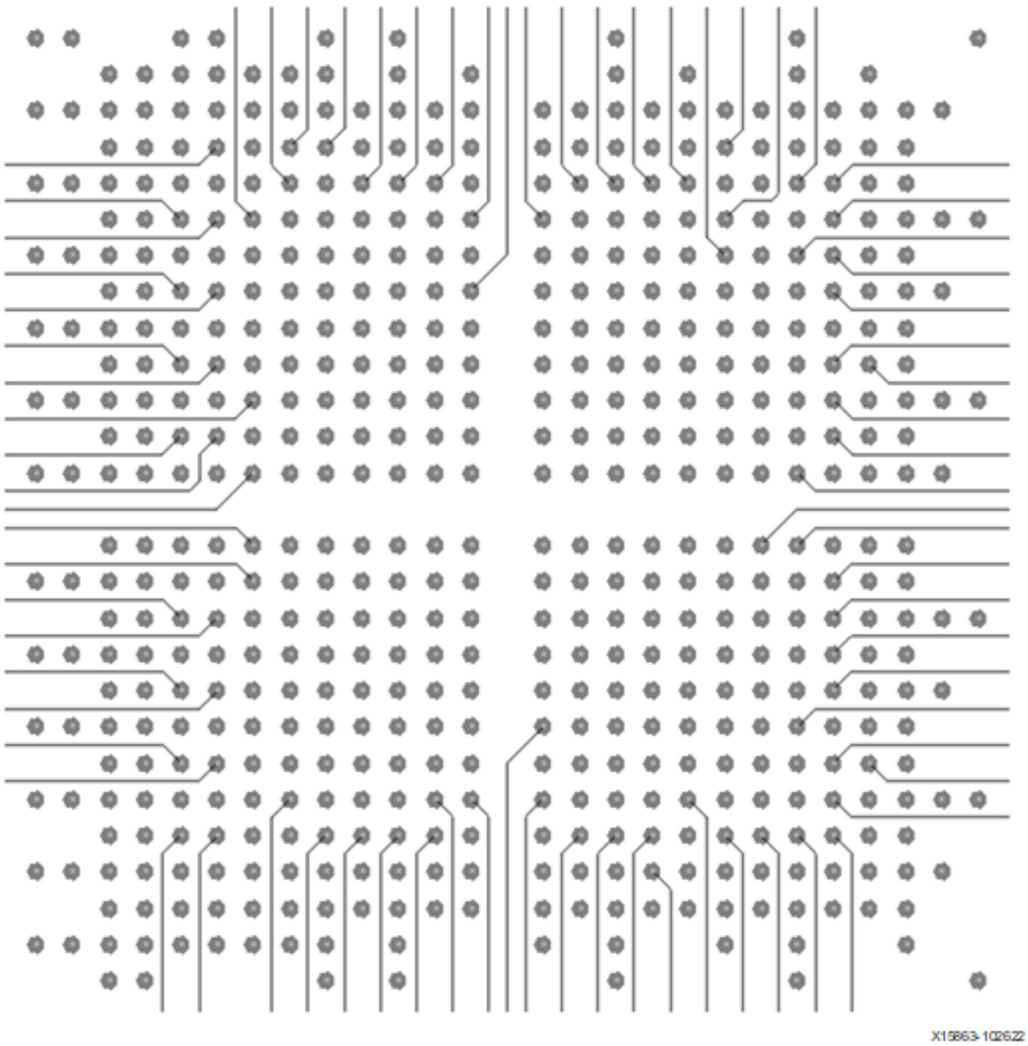


Figure: Inner Signal Layer Three Breakout (1.0 mm/0.92 mm/0.8 mm), One Route between Vias

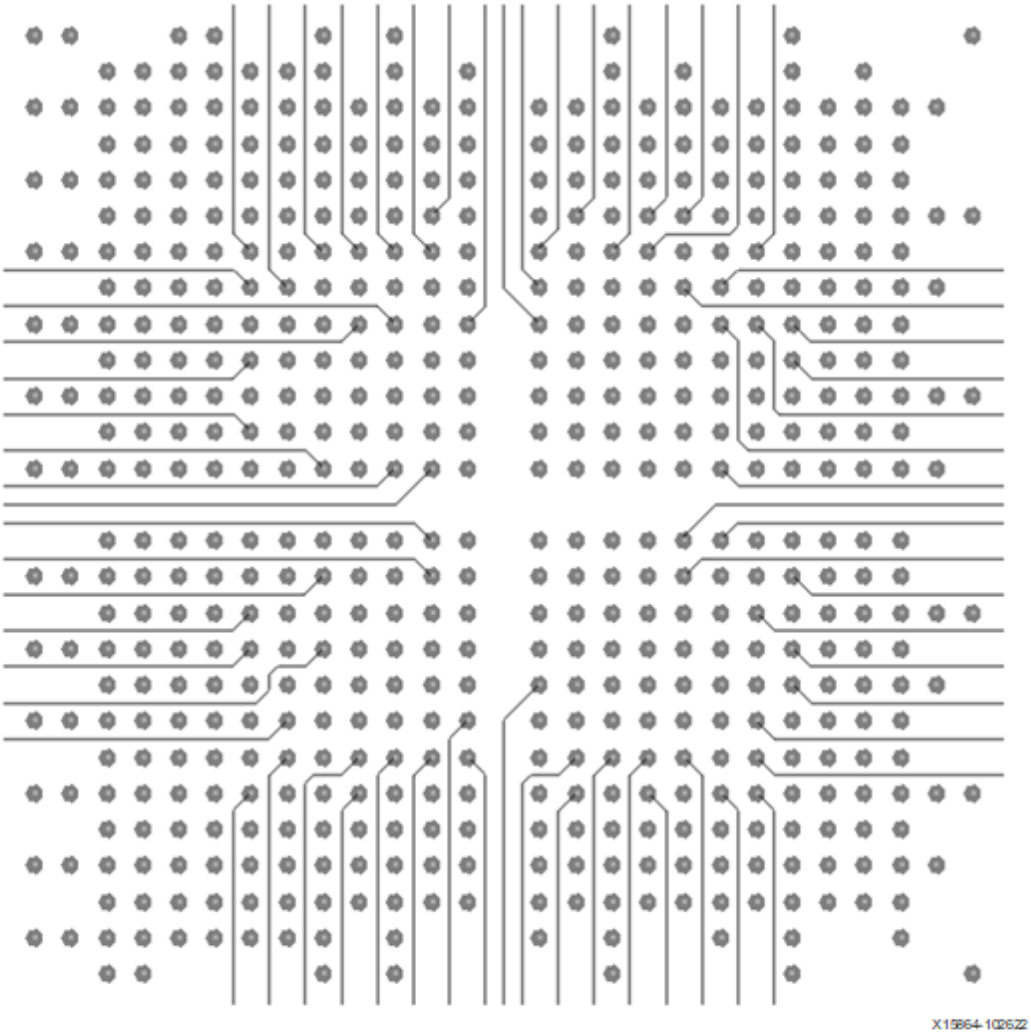
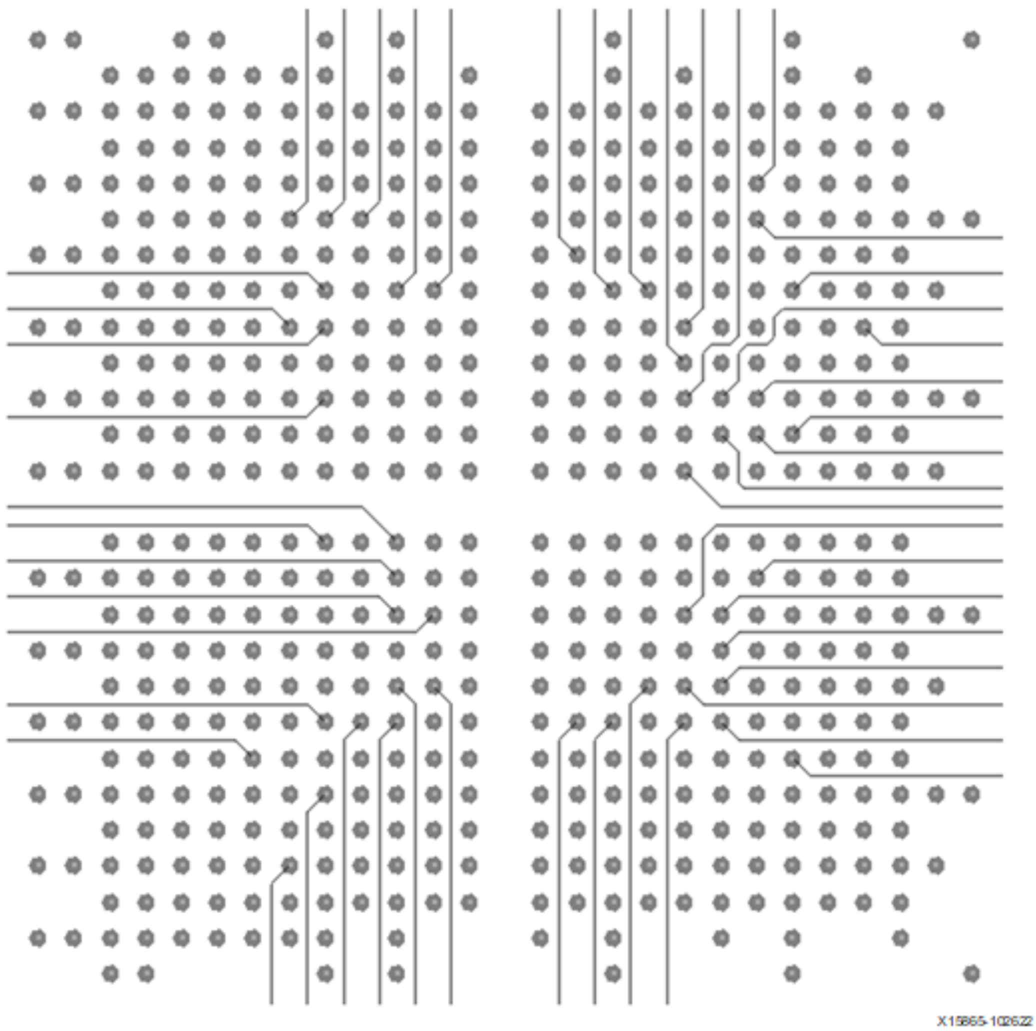


Figure: Inner Signal Layer Five Breakout (1.0 mm/0.92 mm/0.8 mm), One Route between Vias



Sample Breakout with Two Routes between BGA Balls and Vias

Figure: Top Layer Breakout (1.0 mm/0.92 mm/0.8 mm), Two Routes between BGA Balls

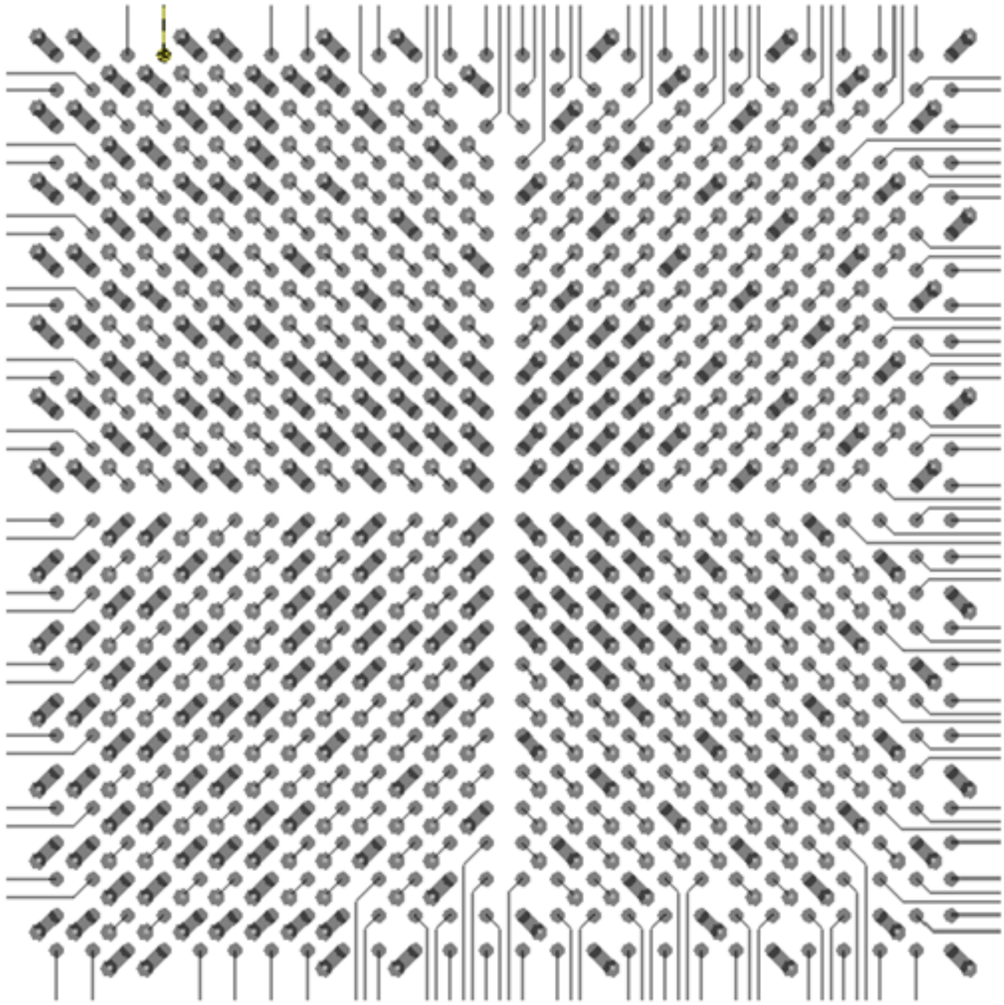


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Figure: Inner Signal Layer One Breakout (1.0 mm/0.92 mm/0.8 mm), Two Routes between Vias

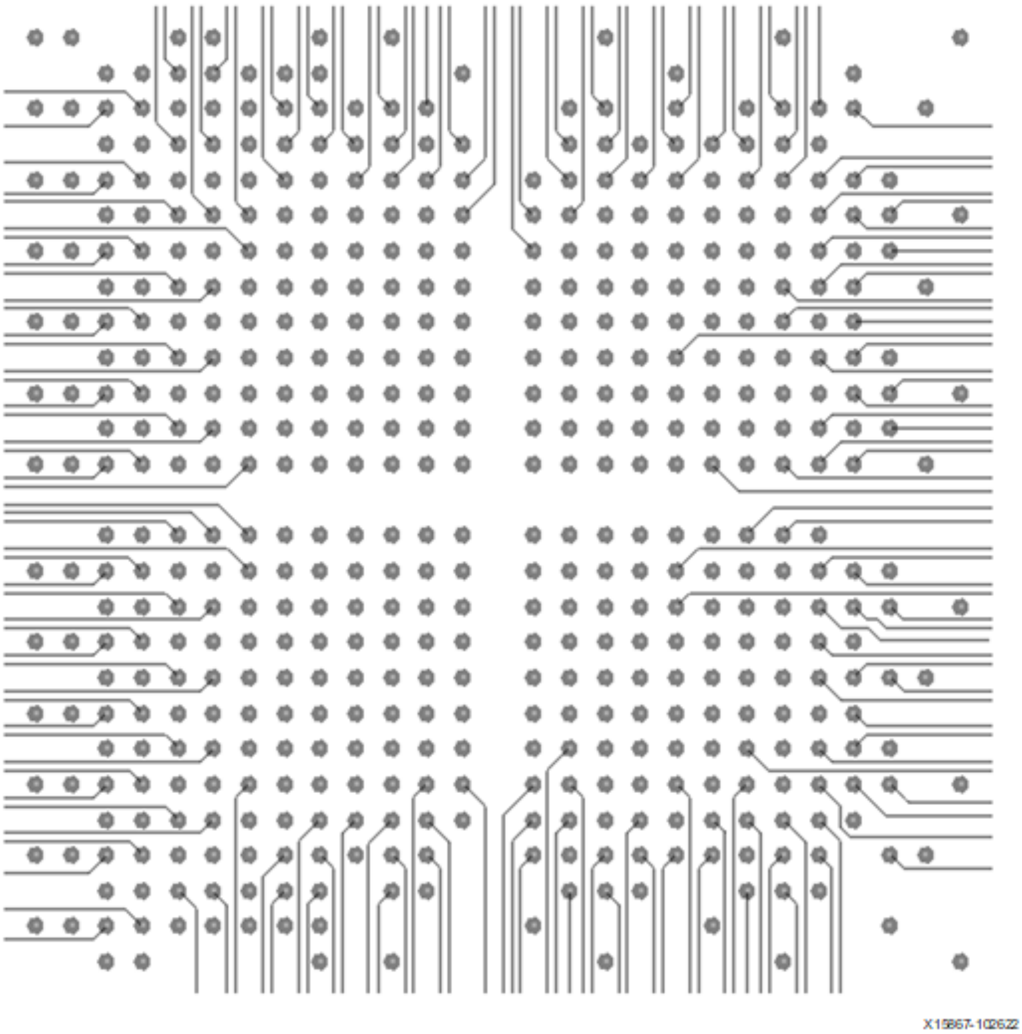
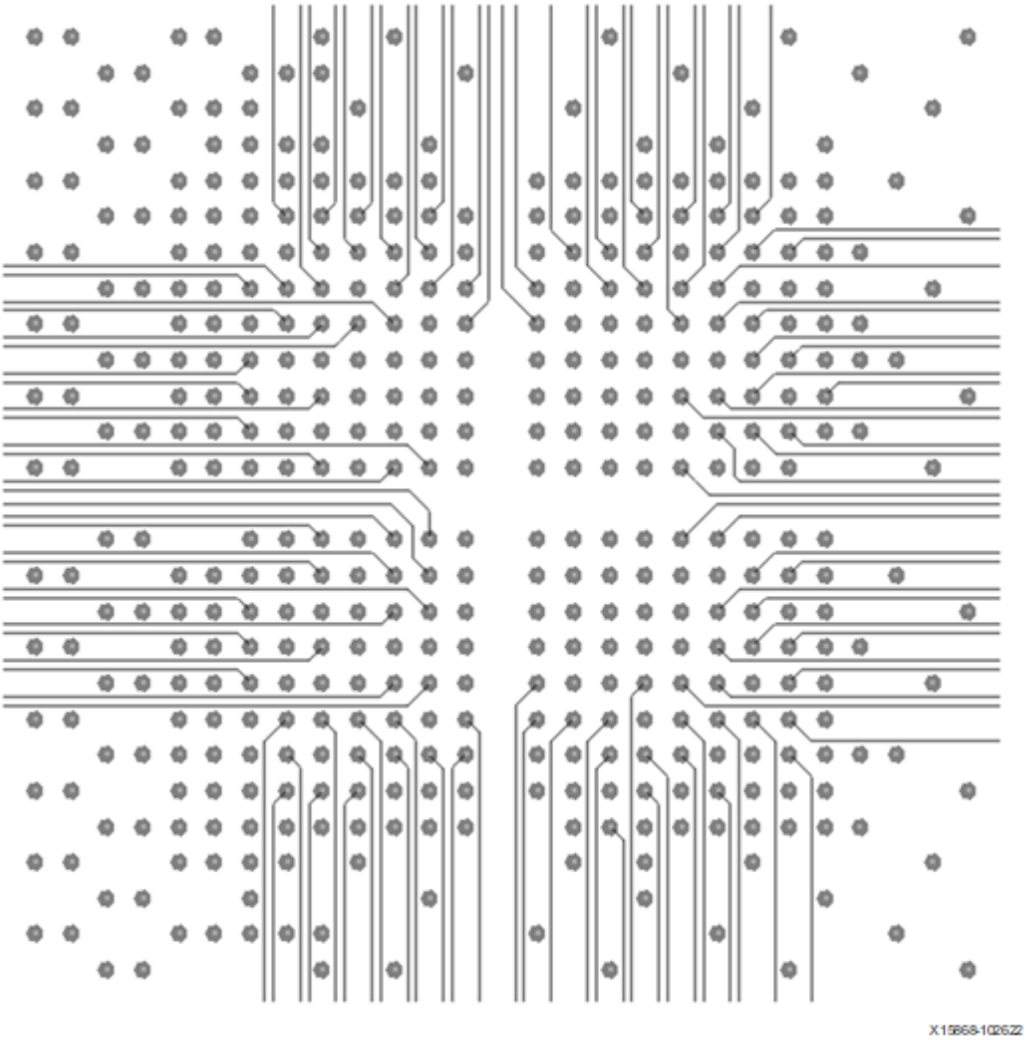


Figure: Inner Signal Layer Two Breakout (1.0 mm/0.92 mm/0.8 mm), Two Routes between Vias



Sample Breakout for 0.5 mm Pitch Devices

The following figures show an example layout showing the BGA routing breakout areas of a 0.5 mm pitch AMD device. The footprint is for a UBVA530 “InFO” device.

Figure: Top Layer Breakout (0.5 mm), One Route between BGA Balls

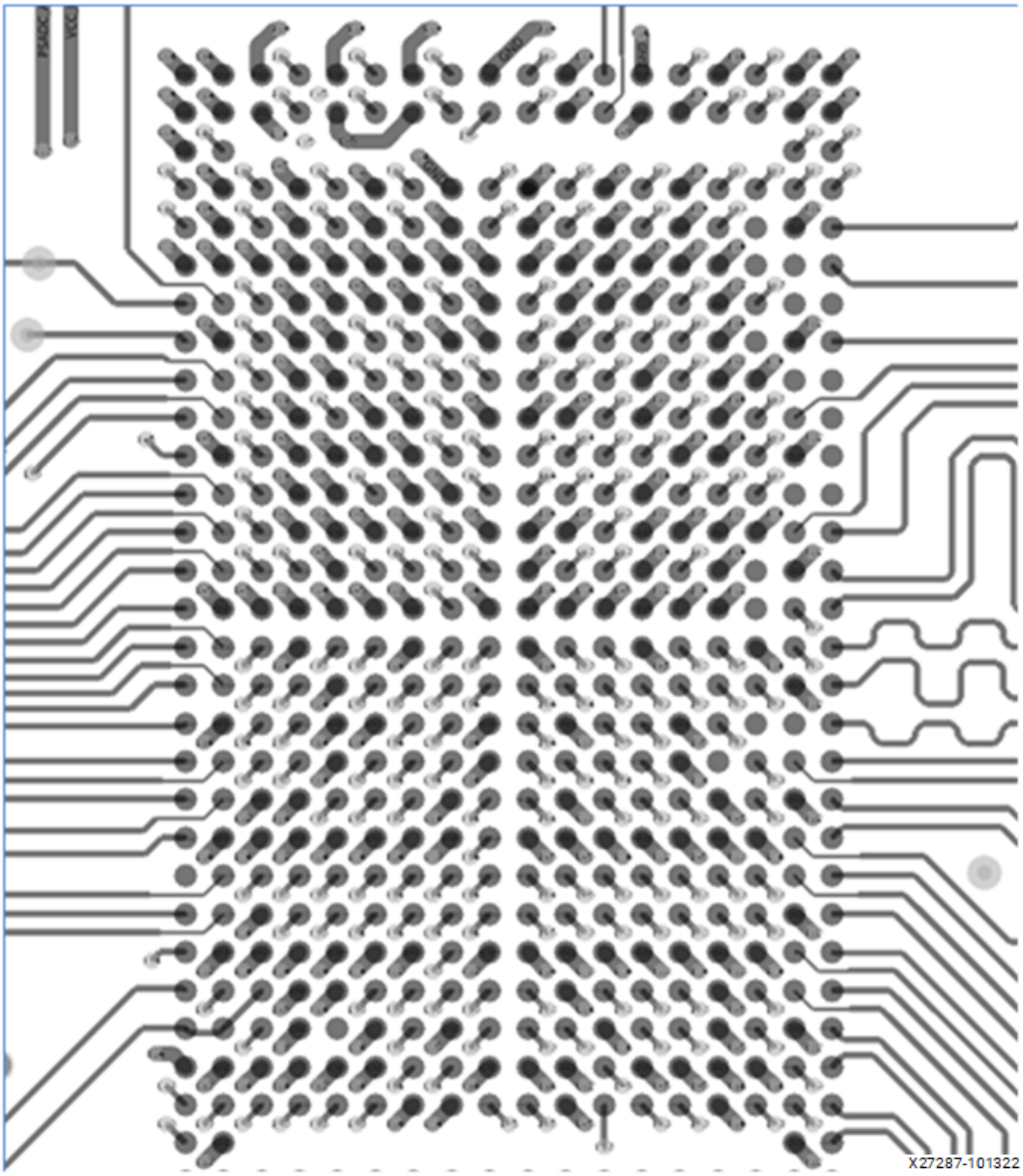


Figure: First Inner Signal Breakout (0.5 mm)

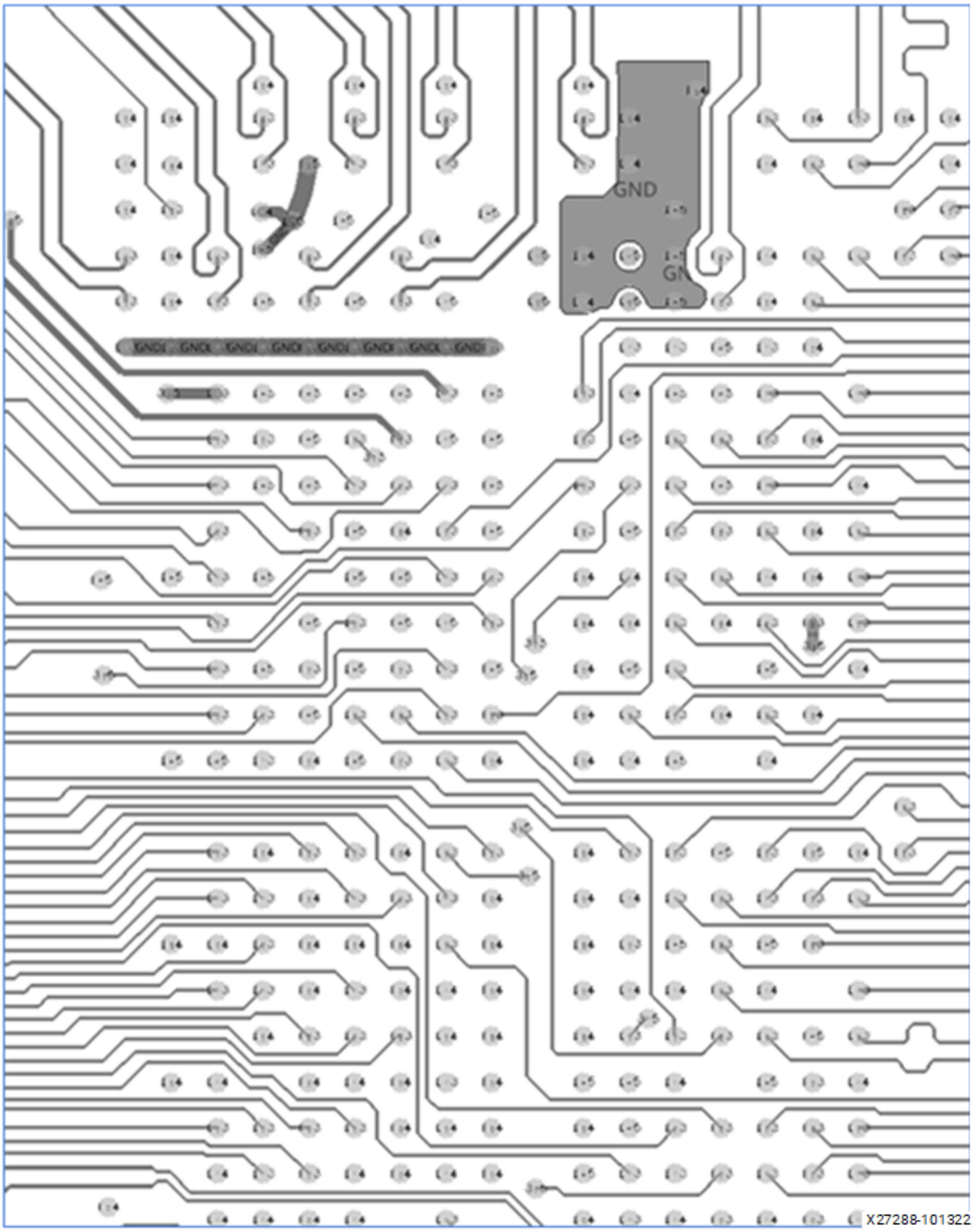
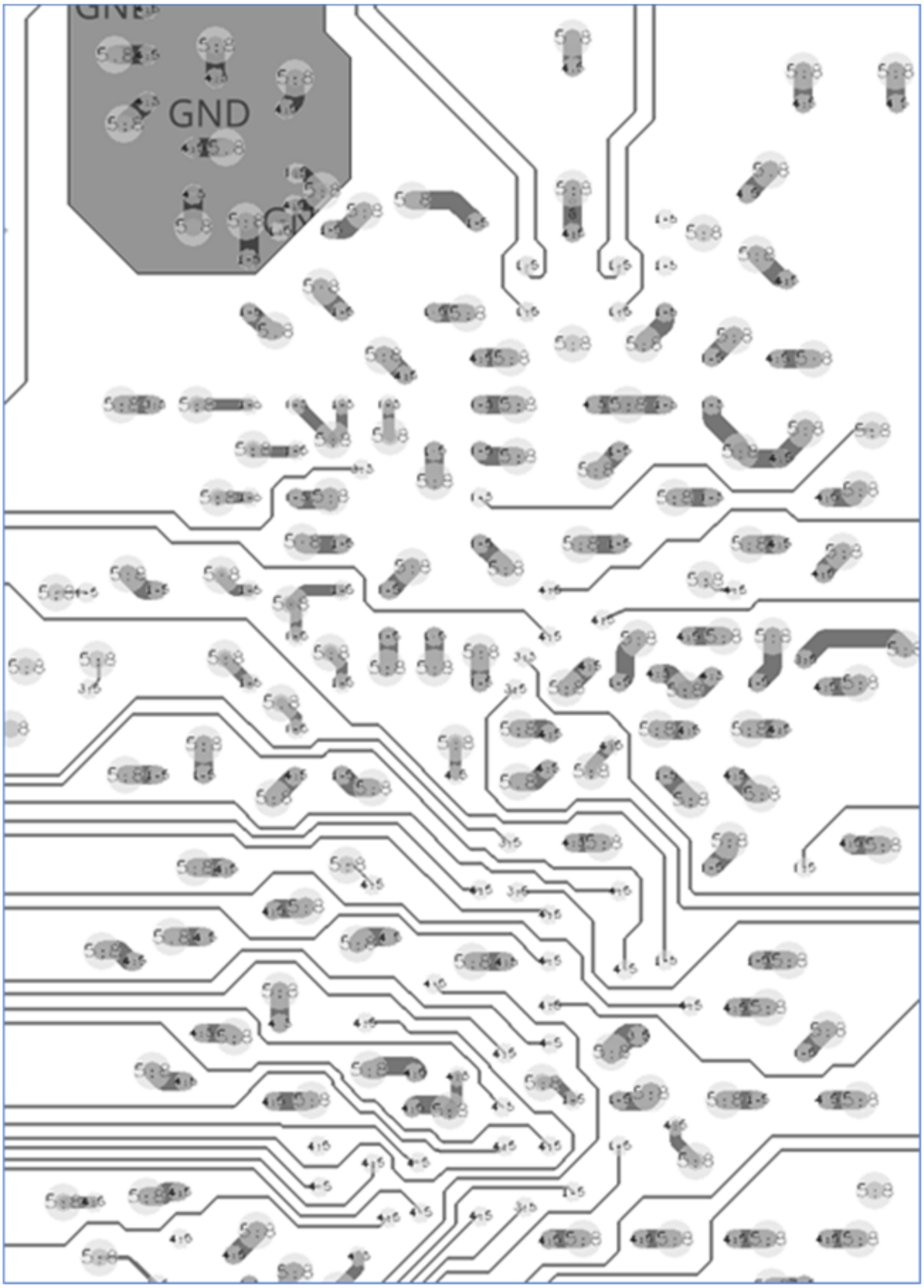


Figure: Second Inner Signal Breakout (0.5 mm)



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